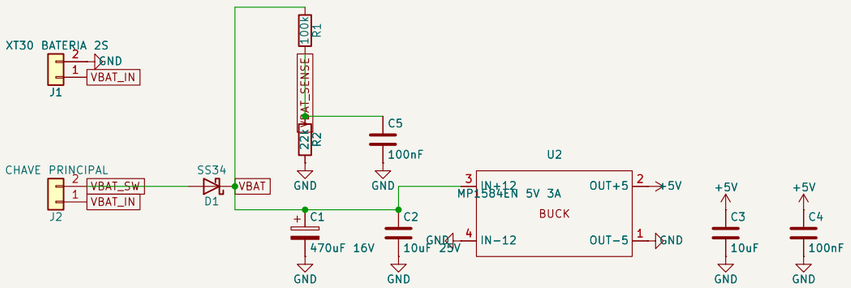
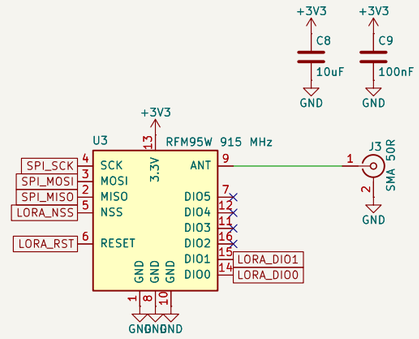
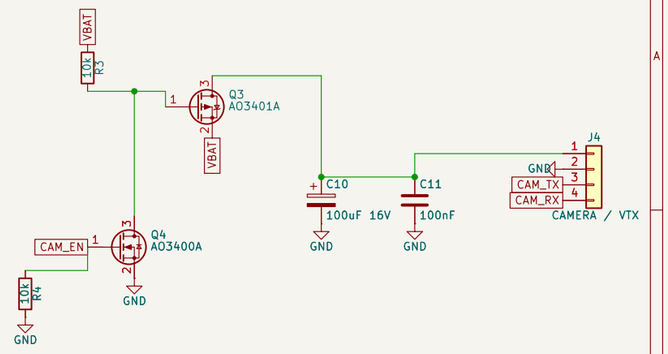
1. ALIMENTAÇÃO E MEDIÇÃO DA BATERIA



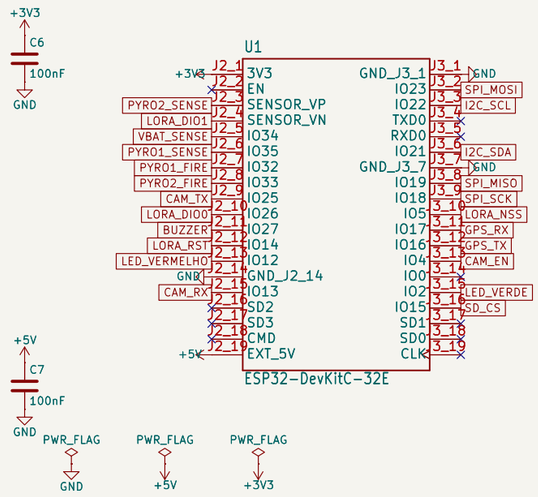
2. TELEMETRIA LoRa 917 MHz



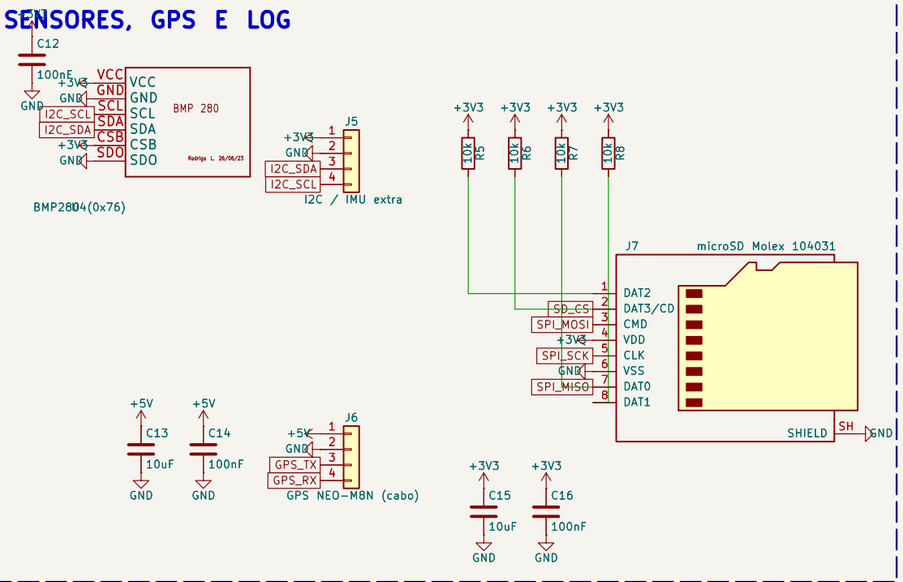
3. CÂMERA / VTX DIGITAL



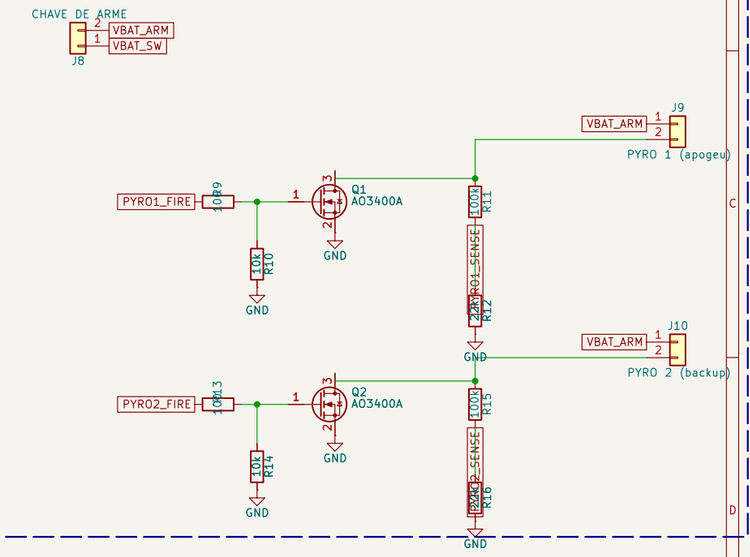
4. MICROCONTROLADOR



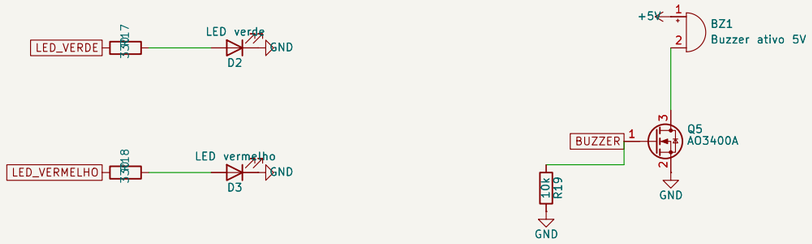
5. SENSORES, GPS E LOG



6. RECUPERAÇÃO (2 CANAIS PIROTÉCNICOS)



7. SINALIZAÇÃO



AVIÔNICA F-500 v2 rev.B — foguete com apogeu ~500 m — UTFPR Rocket — segue o Guia de PCB (Payload 2026)

Corrente de pico (pior caso): ESP32 250 mA + RFM95W 120 mA + microSD 100 mA + GPS 45 mA + buzzer 30 mA + BMP/IMU 6 mA

=> +5 V: ~550 mA (2,75 W); entrada do buck a 80 %: 3,4 W / 6,4 V (2S descarregada) = 0,54 A; câmera DJI O4: até ~1,6 A

=> VBAT ~2,2 A contínuo + pulso pyro ~3 A: trilhas 1,2 mm (Bateria), 1,0 mm (Pyro), 0,6 mm (+5V), 0,5 mm (+3V3), 0,25 mm (dados)

H1 M3 H2 M3 H3 M3 H4 M3

Esquemático em blocos conforme o Guia de PCB (correções Payload 2026)

UTFPR Rocket

Sheet: /

File: avionica_f500_v2.kicad_sch

Title: Aviônica F-500 v2 rev.B — telemetria p/ tracker + câmera

Size: A3	Date: 2026-11-08	**Rev: B**
KiCad E.D.A. 10.0.1		Id: 1/1